

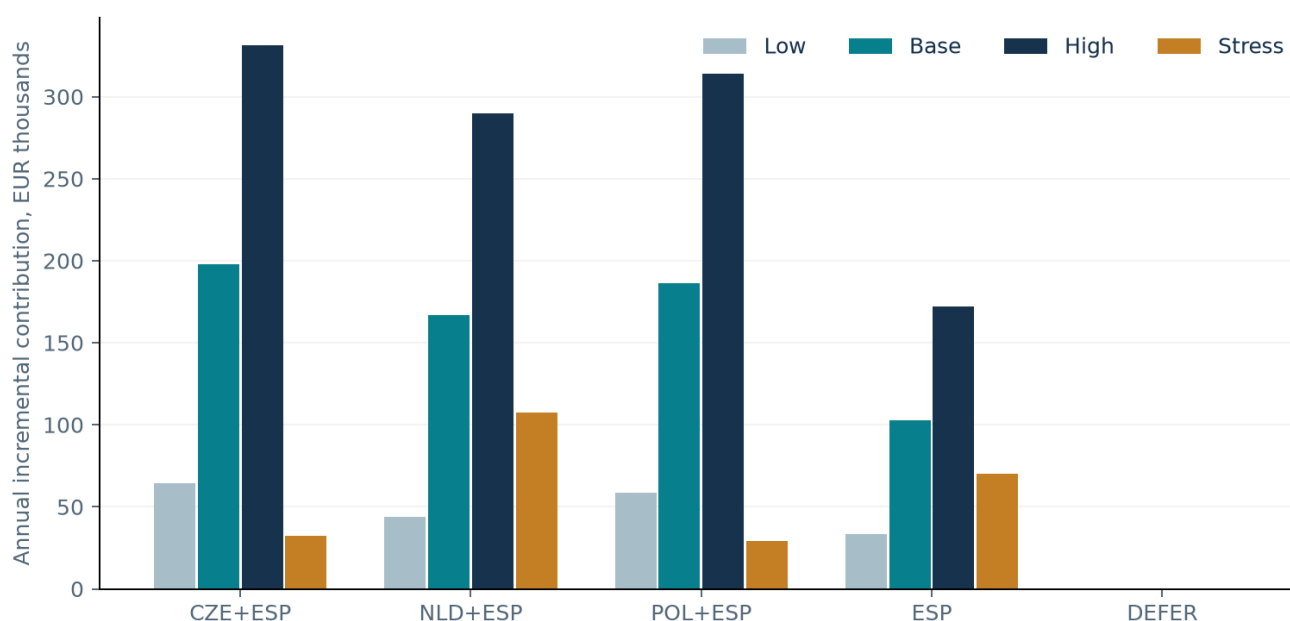
# Fund a staged Czechia and Spain expansion

**Reserve €225,000 and five FTE. Start with Spain; release Czechia after validating demand, saving scope and FX exposure.**

| Annual contribution uplift                     | Year-zero investment                                                                   | Resources                                                     |
|------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Base €198,099<br>Low €64,619<br>Stress €32,180 | €225,000 capex<br>€95,000 recurring fixed cost/year<br>Base simple payback: 1.14 years | 5 of 7 FTE<br>2 of 2 possible hubs<br>€225,000 capex headroom |

**Why this pair.** It produces the highest low, base and high annual increment among all feasible alternatives. Both countries have strong contribution in the reconciled synthetic sales base, and their lower recurring costs support investment at modest volume uplifts. The aggregate specified stress stays positive, although Czechia alone turns negative. Sources: [S06](#), [S08](#), [S09](#), [S11](#), [S12](#), [S14](#).

**The concession.** Netherlands + Spain gives up €31,168 of base annual contribution and uses €45,000 more capex plus one more FTE, but improves the specified stress result by €75,450. It is the preferred alternative if the board prioritizes the worst of the four supplied scenarios over the base case.



**Decision strength: provisional.** These are reproducible scenario results, not a measured effect of opening hubs. Volume uplifts and per-unit savings are synthetic planning assumptions; no probabilities are assigned. A supplemental saving cap combined with stress makes the recommended pair slightly negative (–€1,588/year), reinforcing staged release.

## 2025 OPERATING DIAGNOSIS

## Spain and Czechia lead the contribution pool

The reconciled base contains **1,254 shipped orders and 77,436 units**. Gross sales after discounts are €4,464,622.33; refunds of €80,830.57 reduce net sales to €4,383,791.76. Contribution is €2,030,010.48, or 46.3% of net sales. Sources: [S02](#), [S03](#), [S07](#), [S08](#), [S09](#), [S11](#), [S14](#).

| Market | Orders | Units  | Gross EUR    | Refunds EUR | Net sales EUR |
|--------|--------|--------|--------------|-------------|---------------|
| DEU    | 209    | 12,153 | 631,244.76   | 5,453.00    | 625,791.76    |
| FRA    | 209    | 11,880 | 640,929.94   | 8,699.04    | 632,230.90    |
| NLD    | 209    | 12,576 | 709,108.84   | 11,376.00   | 697,732.84    |
| POL    | 209    | 13,163 | 776,507.79   | 14,617.92   | 761,889.87    |
| CZE    | 209    | 13,733 | 834,572.08   | 19,264.61   | 815,307.47    |
| ESP    | 209    | 13,931 | 872,258.92   | 21,420.00   | 850,838.92    |
| ALL    | 1,254  | 77,436 | 4,464,622.33 | 80,830.57   | 4,383,791.76  |

| Market | Net COGS EUR | Fulfillment EUR | Contribution EUR | Returned units | Margin |
|--------|--------------|-----------------|------------------|----------------|--------|
| DEU    | 349,980.50   | 13,445.70       | 262,365.56       | 73             | 41.9%  |
| FRA    | 341,453.00   | 14,506.80       | 276,271.10       | 144            | 43.7%  |
| NLD    | 365,495.50   | 16,593.12       | 315,644.22       | 144            | 45.2%  |
| POL    | 386,697.50   | 18,698.49       | 356,493.88       | 180            | 46.8%  |
| CZE    | 397,062.00   | 20,910.22       | 397,335.25       | 288            | 48.7%  |
| ESP    | 406,230.50   | 22,707.95       | 421,900.47       | 252            | 49.6%  |
| ALL    | 2,246,919.00 | 106,862.28      | 2,030,010.48     | 1,081          | 46.3%  |

**Operational implication.** Spain and Czechia together generate 40.4% of contribution. Czechia has the highest returned-unit rate (2.10%), followed by Spain (1.81%); validation of returns handling belongs in both pilots. These rates describe physical units, not refund value or the share of orders returned.

Every market has 209 eligible orders. The difference in economics comes from shipped units, price/mix, returns and costs; equal synthetic order counts do not establish equal market opportunity. The workbook contains all 72 country-month records and reconciles them to these totals.

## RECONCILIATION AND ACCOUNTING

## The financial base is a shipment cohort, not cash

| Control               | Observed result                       | Treatment                                                                                                                                |
|-----------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Order reconciliation  | 1,328 raw rows → 1,297 IDs            | 13 identical duplicate rows removed; 18 lower revisions superseded by whole-row corrections.                                             |
| Eligibility           | 1,254 included orders                 | Exclude 24 cancelled, 18 test, one January 2026 shipment. All 24 missing shipment dates belong to cancelled records.                     |
| Valid free shipments  | 12 orders; 739 units                  | Kept in counts and costs; zero sales; combined contribution –€14,565.90.                                                                 |
| Return reconciliation | 264 rows → 243 return IDs             | 20 duplicates removed; one old revision replaced. Include 241 eligible IDs.                                                              |
| Cutoff and orphan     | R-CUTOFF included; two excluded       | 2026-01-31 is inclusive. R-LATE (2026-02-01; €33) excluded. R-ORPHAN (2 units, 60 local) quarantined; currency unknown.                  |
| COGS recovery         | 481 restocked / 1,081 returned units  | Recover €18,485.50 at original shipment cost; no cost recovery for 600 un-restocked units.                                               |
| Missingness / joins   | No eligible monetary, FX or cost gaps | No imputations; no conflicting equal revisions. No observed eligible order has multiple distinct return IDs; code supports additive IDs. |

**Per-order accounting.** Gross local = units × local price – discount. Gross EUR and summed order credits are each divided by the original shipment month’s ECB mean. Gross COGS and recovered COGS use the latest effective SKU cost on or before shipment. Each monetary component is rounded half-up to cents before aggregation. Fulfillment is nonrefundable. Net sales = gross – refunds; contribution = net sales – net COGS – fulfillment. Margin is null when net sales is zero. Sources: [S02](#), [S03](#), [S14](#).

**Boundary examples.** R-CUTOFF is received 31 January 2026 but reduces the December 2025 shipment cohort by €54 and recovers €29.50 of BRAKE cost. R-M1-01-003 revision 2 replaces the €150 credit with €149. The future order FUTURE-2026 does not enter 2025 even though its date is before the return cutoff. Sources: [S07](#), [S09](#), [S11](#).

**Booked revenue, cash and contribution.** This report’s sales measure is a reconstructed tax-exclusive shipped-sales cohort, net of known credits, not a general-ledger audit. Credit timing is assigned back to the shipment month, so monthly results need not equal a calendar-month revenue ledger. Cash depends on collection/refund dates, credit terms, taxes and realized FX; none is supplied. Contribution further deducts product and fulfillment costs, but is before existing central overhead, tax, financing and hub capex.

ARCHIVED WORLD BANK OBSERVATIONS

## Public market context informs scale, not product demand

Germany and France have the largest populations; the Netherlands has the highest current-US-dollar GDP per capita. Poland’s population declined while the other five grew from 2022 to 2024. These observations describe national context. They neither measure replacement-assembly demand nor explain the synthetic sales ranking. Sources: [S05](#), [S10](#), [S13](#).

| Market | Population 2022 | Population 2023 | Population 2024 | 2022-24 change |
|--------|-----------------|-----------------|-----------------|----------------|
| DEU    | 83,177,813      | 83,287,273      | 83,516,593      | 0.41%          |
| FRA    | 68,184,457      | 68,372,286      | 68,551,653      | 0.54%          |
| NLD    | 17,700,982      | 17,877,117      | 17,993,485      | 1.65%          |
| POL    | 36,821,749      | 36,687,353      | 36,559,233      | -0.71%         |
| CZE    | 10,672,118      | 10,864,042      | 10,905,028      | 2.18%          |
| ESP    | 47,786,102      | 48,352,528      | 48,848,840      | 2.22%          |

| Market | GDP/person 2022 | GDP/person 2023 | GDP/person 2024 |
|--------|-----------------|-----------------|-----------------|
| DEU    | \$50,506.52     | \$54,776.77     | \$56,103.73     |
| FRA    | \$40,988.64     | \$44,700.14     | \$46,103.08     |
| NLD    | \$59,123.32     | \$63,515.60     | \$67,465.32     |
| POL    | \$18,891.21     | \$22,145.27     | \$25,103.57     |
| CZE    | \$28,282.22     | \$31,761.59     | \$31,827.76     |
| ESP    | \$30,319.50     | \$33,493.22     | \$35,326.77     |

**Interpretation limits.** GDP per capita is current US\$, not purchasing-power parity or constant-price real income. Changes include price and exchange-rate effects and are not evidence of equivalent real purchasing-power growth. Population differences do not proxy installed equipment, channel access or service intensity. No market-size or causal uplift estimate is derived from these series.

**Vintage and completeness.** Both World Bank responses contain all 18 requested country-years with no null values. Original API pagination reports one page; source lastupdated is 2026-07-13. The archive fetched them on 2026-09-27 at 06:36 UTC; this analysis collected those saved bytes at 10:12 UTC the same day. Historical values may incorporate later revisions. The workbook and JSON preserve full source precision; this page displays GDP per capita to cents.

**Next decision evidence.** Commercial leads should validate serviceable installed base, customer service needs and qualified pipeline by country. No customer interviews, competitor study or live demand research were performed. The transaction base is synthetic, independently of the public source status.

## SINGLE-HUB ECONOMICS

## Lower fixed cost makes Czechia and Spain attractive

**Normal case:**  $\Delta C = C \times u + U \times (1 + u) \times s - F$ . Here C is 2025 contribution, U shipped units, s saving per unit, F recurring annual fixed cost, and u is 10% / 25% / 40% for low / base / high. **Stress:**  $\Delta C = (C - 0.03G - FX \text{ shock}) \times 1.25 - C + U \times 1.25 \times s - F$ ; FX shock = 0.10N in PLN/CZK markets only. G and N are gross and net sales. Sources: [S06](#), [S12](#).

| Market | Capex EUR | FTE | Fixed EUR/yr | Saving EUR/unit | Break-even uplift | Base payback years |
|--------|-----------|-----|--------------|-----------------|-------------------|--------------------|
| DEU    | 280,000   | 5   | 95,000       | 2.60            | 21.57%            | 27.75              |
| FRA    | 210,000   | 4   | 85,000       | 2.90            | 16.27%            | 7.74               |
| NLD    | 140,000   | 3   | 65,000       | 3.20            | 6.96%             | 2.18               |
| POL    | 110,000   | 3   | 45,000       | 2.40            | 3.46%             | 1.32               |
| CZE    | 95,000    | 2   | 40,000       | 2.10            | 2.62%             | 1.00               |
| ESP    | 130,000   | 3   | 55,000       | 3.00            | 2.85%             | 1.27               |

| Market | Low $\Delta C$ EUR | Base $\Delta C$ EUR | High $\Delta C$ EUR | Stress $\Delta C$ EUR |
|--------|--------------------|---------------------|---------------------|-----------------------|
| DEU    | -34,005.86         | 10,088.64           | 54,183.14           | -13,583.04            |
| FRA    | -19,475.69         | 27,132.78           | 73,741.24           | 3,097.90              |
| NLD    | 10,831.94          | 64,215.06           | 117,598.17          | 37,623.47             |
| POL    | 25,399.71          | 83,612.47           | 141,825.23          | -40,742.81            |
| CZE    | 31,456.76          | 95,382.94           | 159,309.12          | -37,826.95            |
| ESP    | 33,162.35          | 102,716.37          | 172,270.39          | 70,006.66             |

Spain has the highest base increment (€102,716), with Czechia second (€95,383). Germany requires a 21.57% volume uplift to break even and has a 27.75-year base payback; France also loses contribution in the low case. They are unattractive priorities despite larger national populations.

**Timing matters.** Capex is a year-zero investment; the table shows a full year at steady state after recurring fixed cost. Simple payback is capex divided by positive annual increment and is null when that increment is zero or negative. Payback excludes ramp-up, financing, tax, working capital and residual value. A 90-day implementation is not an assumption that the first year achieves a full run rate.

**Stress is deliberately conservative.** It compares the shocked hub case with an unchanged normal baseline, per policy. It is not a forecast probability, a mechanically recalculated 10% exchange quote depreciation, or a comparison of both alternatives under the same macro shock. Pair benefits and resources add without synergy. Supplemental analyses do not replace this required definition.

## FEASIBLE SET AND RANKINGS

## All feasible portfolios have been tested

| Base rank | Portfolio | Capex €k | FTE | Low €k/yr | Base €k/yr | High €k/yr | Stress €k/yr | Base PB yrs |
|-----------|-----------|----------|-----|-----------|------------|------------|--------------|-------------|
| 1         | CZE+ESP   | 225      | 5   | 64.6      | 198.1      | 331.6      | 32.2         | 1.14        |
| 2         | POL+ESP   | 240      | 6   | 58.6      | 186.3      | 314.1      | 29.3         | 1.29        |
| 3         | POL+CZE   | 205      | 5   | 56.9      | 179.0      | 301.1      | -78.6        | 1.15        |
| 4         | NLD+ESP   | 270      | 6   | 44.0      | 166.9      | 289.9      | 107.6        | 1.62        |
| 5         | NLD+CZE   | 235      | 5   | 42.3      | 159.6      | 276.9      | -0.2         | 1.47        |
| 6         | NLD+POL   | 250      | 6   | 36.2      | 147.8      | 259.4      | -3.1         | 1.69        |
| 7         | FRA+ESP   | 340      | 7   | 13.7      | 129.8      | 246.0      | 73.1         | 2.62        |
| 8         | FRA+CZE   | 305      | 6   | 12.0      | 122.5      | 233.1      | -34.7        | 2.49        |
| 9         | FRA+POL   | 320      | 7   | 5.9       | 110.7      | 215.6      | -37.6        | 2.89        |
| 10        | DEU+CZE   | 375      | 7   | -2.5      | 105.5      | 213.5      | -51.4        | 3.56        |
| 11        | ESP       | 130      | 3   | 33.2      | 102.7      | 172.3      | 70.0         | 1.27        |
| 12        | CZE       | 95       | 2   | 31.5      | 95.4       | 159.3      | -37.8        | 1.00        |
| 13        | FRA+NLD   | 350      | 7   | -8.6      | 91.3       | 191.3      | 40.7         | 3.83        |
| 14        | POL       | 110      | 3   | 25.4      | 83.6       | 141.8      | -40.7        | 1.32        |
| 15        | NLD       | 140      | 3   | 10.8      | 64.2       | 117.6      | 37.6         | 2.18        |
| 16        | FRA       | 210      | 4   | -19.5     | 27.1       | 73.7       | 3.1          | 7.74        |
| 17        | DEU       | 280      | 5   | -34.0     | 10.1       | 54.2       | -13.6        | 27.75       |
| 18        | DEFER     | 0        | 0   | 0.0       | 0.0        | 0.0        | 0.0          | —           |

**Hard constraints, supplied by the board:** no more than €450,000 capex, seven FTE and two hubs. Eleven of 15 possible pairs are feasible. DEU+FRA fails both (€490,000; 9 FTE). DEU+NLD (€420,000), DEU+POL (€390,000) and DEU+ESP (€410,000) each need 8 FTE and fail staffing. All six singles and defer are feasible. Sources: [S06](#), [S12](#).

**Decision rule, consultant assumption:** prioritize base annual increment under hard limits, check downside and stage spend; no weighted score or scenario probabilities. CZE+ESP dominates POL+ESP on all four contribution cases, capex and FTE. NLD+ESP is incomparable: less base upside and more resources, but stronger stress. Under a maximin rule across the four supplied cases, NLD+ESP wins with a minimum €43,994, versus €32,180 for CZE+ESP.

Table values are € thousands rounded for legibility; the workbook and metrics.json retain cents and payback for every case, including infeasible alternatives. Defer is zero incremental contribution under the policy benchmark; it retains current operations.

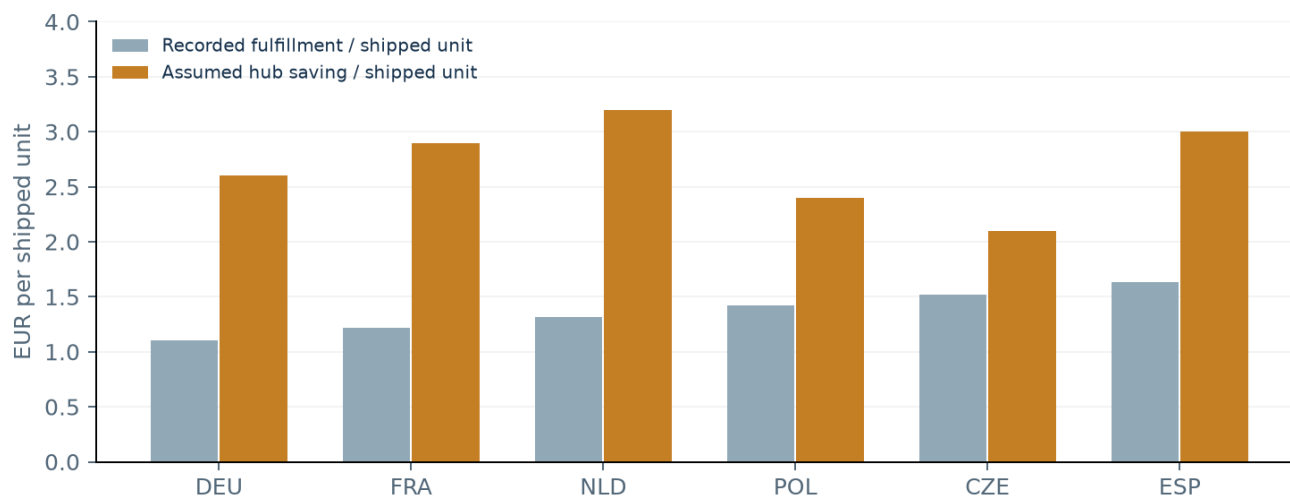
## SWITCHING ASSUMPTIONS AND RISKS

## The second hub should earn its release

| Comparison                                | CZE + ESP    | NLD + ESP    | What is given up                           |
|-------------------------------------------|--------------|--------------|--------------------------------------------|
| Base annual increment                     | €198,099     | €166,931     | Choose NLD: –€31,168/year                  |
| Specified stress                          | €32,180      | €107,630     | Choose CZE: –€75,450/year in stress        |
| Capex / FTE                               | €225,000 / 5 | €270,000 / 6 | Choose NLD: +€45,000 and +1 FTE            |
| Cap savings at recorded fulfillment: base | €164,332     | €113,513     | Analyst conservative saving boundary       |
| Cap savings + specified stress            | €-1,588      | €54,211      | Recommended pair becomes slightly negative |

**Demand flip.** With the Netherlands held at 25% uplift and normal FX/returns, Czechia must deliver about **17.7%** uplift to match its annual increment. Czechia's own normal-case operating break-even is only 2.62%, but merely clearing break-even does not make it the best second hub. The recommended pair's common uplift break-even is 2.74%; at zero uplift it loses €24,368/year.

**FX flip.** At base volume and an additional refund cost of 3% of gross sales in both markets, Netherlands overtakes Czechia once the Czech net-sales FX haircut exceeds approximately **2.6%**. Czechia's own increment reaches zero at about a 6.3% haircut. These are model sensitivities using the policy's revenue-haircut convention, not forecast exchange rates or hedging advice.



**Validate the saving scope first.** Czechia's assumed €2.10/unit exceeds recorded fulfillment of €1.52; Spain's €3.00 exceeds €1.63. Every option has this mismatch. Savings may involve other costs, but their scope is unspecified. The cap sensitivity assumes fulfillment is the only pool, and even full elimination is an optimistic operational bound. Procurement and Finance must obtain a net saving bridge, including local variable costs and any stranded central cost, before release. Sources: [S06](#), [S08](#), [S09](#), [S12](#).

## EXECUTION PROPOSAL — ROLES AND DATES ARE PLANNING ASSUMPTIONS

## A 90-day plan with separate capital gates

| Stage / owner                                          | Work and dependency                                                                                                                                                                                                                            | Measurable gate / decision                                                                                                                                                                                   |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Days 1–15<br><b>CFO + data lead</b>                    | Reconcile ledger to the saved cohort; resolve orphan documentation; establish SKU/customer mix and service baseline. Treasurer maps CZK receivables and EUR cost exposure. Depends on ledger/payment exports and service-event definitions.    | Gate A: no unexplained financial variance; owner assigned to each return exception. Produce all-in capex and working-capital estimate. Confirm €225k envelope / 5 FTE remains viable.                        |
| Days 16–30<br><b>COO + Procurement + country leads</b> | Obtain site and service quotes; document net unit-saving bridge; qualify Spanish and Czech pipeline. Select comparison customers served centrally. HR validates service-window coverage. Depends on demand/quote inputs from Gate A.           | Gate B: release Spain within €130k and 3 FTE only if low case stays positive using validated net savings, and supply/IT readiness is complete. Reserve Czechia €95k / 2 FTE; do not bind its site yet.       |
| Days 31–60<br><b>Spain operations lead + IT + HR</b>   | Launch a limited SKU/service pilot after integration, stock traceability, staff training and fulfillment acceptance. Keep a matched central-service comparison and weekly mix-adjusted reporting. Depends on Gate B and stocked eligible SKUs. | Gate C: ≥95% eligible pilot orders dispatched within one business day; ≥99% inventory accuracy; no critical unresolved reconciliation defect. Four weekly reviews of cost/unit, contribution and exceptions. |
| Days 61–75<br><b>CFO + Commercial + Treasury</b>       | Update scenarios with pilot unit economics, season-matched demand and receivables FX exposure. Refresh Netherlands alternative with equally scoped quotes. Wait for return cohorts to mature where needed.                                     | Gate D: choose CZE, NLD or defer second hub. Reopen CZE below ~17.7% uplift if NLD remains 25%, or above ~2.6% FX haircut with 3% refund shock. Recalculate both thresholds if savings/costs change.         |
| Days 76–90<br><b>COO + board sponsor</b>               | If Gate D passes, contract and prepare second hub with limited stock and staff; otherwise keep Spain only and review next quarter. An NLD substitution needs a revised €270k / 6 FTE envelope.                                                 | Gate E: confirm projected positive annual increment, resource compliance and KPI ownership. Report committed capex separately from annualized benefit; no automatic second-site launch date.                 |

**Staffing concept.** Spain: one site lead, one parts/dispatch coordinator and one service technician (3 FTE). Czechia: one lead technician and one stock/dispatch coordinator (2 FTE). These are proposed allocations, not a validated shift roster; HR must test absence coverage and opening hours. The remaining two FTE and €225,000 are headroom, not committed reserve spending.

**Readiness limitation.** Local permits, lease terms, inventory investment, IT effort and recruitment lead times are unpriced. Existing owners' project time is not automatically included in hub FTE or capex. Confirm these costs at Gate A; do not assert an operational launch within 90 days if dependencies fail.

## KPI AND RISK PLAN

## Measure contribution, service and capital separately

| KPI / owner                                  | Baseline / proposed target                                                                                              | Measurement and cadence                                                                                                                                        |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incremental contribution<br>CFO              | Model base: CZE €95,383; ESP €102,716 annually. No measured uplift. Target: positive validated low case before release. | Monthly bridge: mix-adjusted net sales – net product/fulfillment cost – hub fixed cost versus central comparison. Report run rate and realized YTD separately. |
| Volume uplift<br>Commercial leads            | Base assumption 25%; low 10%. CZE selection threshold ~17.7% vs NLD at 25%.                                             | Weekly shipped units vs same 2025 seasonal period and matched central-service cohort; exclude test/future orders and log price/mix changes.                    |
| Net saving per unit<br>Procurement / Finance | Assumed CZE €2.10; ESP €3.00. Validate rather than treat as achieved.                                                   | Weekly cost bridge for transport, handling, product and local variable costs; reconcile invoices monthly. No double-counting COGS recovery.                    |
| Returned-unit rate / refunds<br>Quality lead | 2025: CZE 2.10%, ESP 1.81% units. Proposed alert: >baseline +0.5 percentage points.                                     | Monthly shipment cohorts, age-matched; show refund/gross sales and reusable share separately. Proposed alert is not the policy's 3% gross-sales credit shock.  |
| Service / stock accuracy<br>Operations lead  | Baseline unavailable. Proposed: ≥95% dispatched within one business day; ≥99% stock accuracy.                           | Timestamp eligible orders and dispatch; weekly service dashboard, weekly cycle counts. Validate customer relevance before setting service commitments.         |
| Capex, fixed cost and staffing<br>CFO / HR   | CZE+ESP: capex ≤€225k, fixed cost €95k/year, 5 FTE. Absolute board ceilings €450k / 7 FTE.                              | Weekly purchase-commitment log and monthly P&L/payroll review. Forecast deposits, inventory and implementation cash separately.                                |

**Risk priorities.** Demand attribution and saving scope are high-priority unknowns: control comparisons and invoice-based cost bridges address them. FX can reverse the second-site choice: Treasury updates the exposure sensitivity monthly. Return censoring can overstate margins: Quality tracks mature cohorts and separates reusable from non-reusable units. Site/ramp costs can extend payback: Finance gates binding spend on an all-in cash plan.

The pilot can establish readiness and improve unit-cost estimates; a short, small sample cannot prove a sustained 25% demand lift. No statistical power claim is made. If return maturity or seasonality prevents a fair read by day 75, retain the Spain-only stage and defer the second-site commitment.

## CLAIM-TO-SOURCE ASSESSMENT

## What the evidence supports—and what remains unverified

**Claim and threshold.** For these six markets, the predictive claim is that a hub can generate incremental annual contribution sufficient to justify investment. The decision threshold is positive modeled contribution within the capex/FTE constraints; preference for base-case maximization is a consultant default. Calculation accuracy and real-world hub impact are separate claims.

| Claim / evidence state                                                         | Basis, directness and quality                                                                                                                                                                                           | Conclusion / reopening evidence                                                                                                                                               |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 contribution ranking<br><b>Computed / supported within synthetic case</b> | S02, S07–S09, S11, S14: transaction exports and explicit accounting rules; complete downloaded pages, controlled IDs/revisions, effective cost schedule. Direct for the defined synthetic cohort; no ledger/cash audit. | CZE and ESP have the strongest contribution pools. Reopen on valid source corrections, cost changes or eligible credits received by cutoff.                                   |
| National context<br><b>Verified archive bytes / contextual</b>                 | S05, S10, S13: official aggregate series, all 18 country-years; hashes and metadata checked. Strong for national measures, indirect for the product; later historical revisions possible.                               | Preserve population and current-US\$ income context. Demand claim remains unverified. Reopen context on a documented source revision; demand needs product/customer evidence. |
| Historical FX translation<br><b>Computed / supported</b>                       | S03–S04: official reference observations; 255 published dates per required currency, 12 monthly means; ZIP extraction verified. Valid translation convention, not realized FX.                                          | Use shipment-month quote for sale/refund. Reopen on archived observation changes or an explicit alternative accounting convention.                                            |
| Hub uplift and savings<br><b>Assumed / unverified effect</b>                   | S06, S12: six synthetic options and scenario ranges. Direct to the requested arithmetic; no experiment, quote support, customer research or reference-case validation. Saving scope is unresolved.                      | Hold any claim that hubs cause 25% growth. Validate demand and net savings; second-hub choice flips at the quantified thresholds.                                             |
| Staged CZE+ESP decision<br><b>Inferred / provisional</b>                       | Reproducible portfolio comparison plus explicit consultant preference; no probabilities, invented weights or claim of certainty. NLD+ESP remains stronger in stress.                                                    | Reserve the envelope and test before committing. Maximin preference, weaker Czech demand, FX exposure or saving revisions can change the choice.                              |

**Quality dimensions.** Transaction completeness is supported within the supplied exports; measurement validity is bounded by the dictionary and cutoff, and synthetic generation limits external applicability. Official series have strong provenance for their stated units, but limited product relevance. Scenario design is transparent; causal rigor is absent, optimism/selection bias is possible, and implementation cash costs are incomplete. These qualitative judgments do not form a composite score.

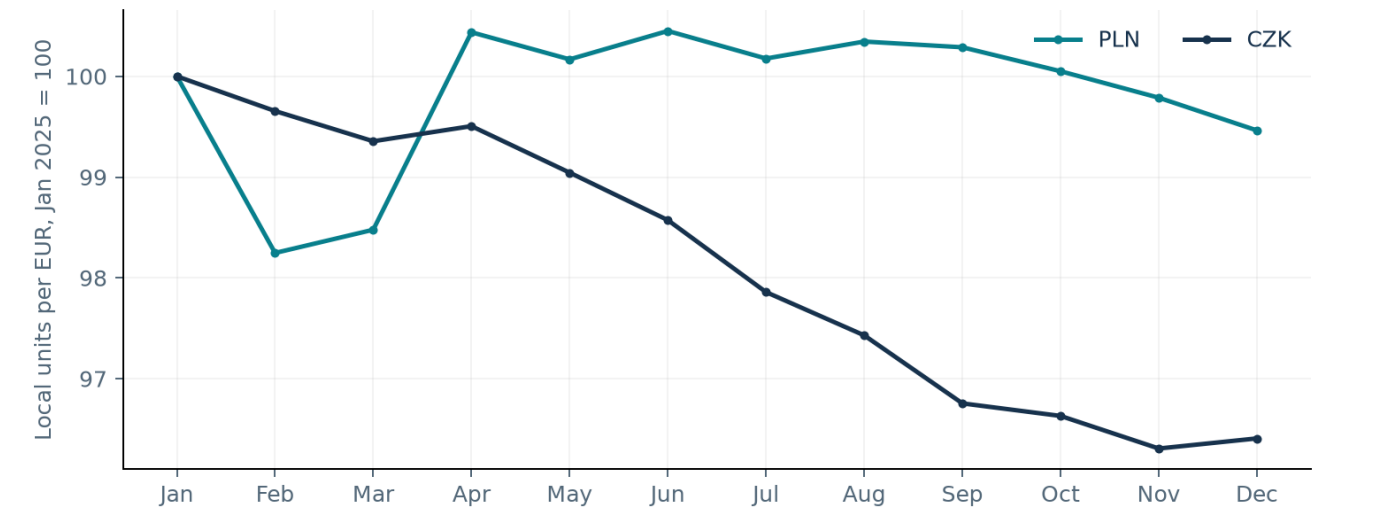
**Agreement and conflicts.** Official and client sources measure different things and do not independently corroborate hub demand. The required model reconciles internally. The saving-versus-fulfillment gap is an unresolved scope issue, not permission to alter supplied assumptions silently. Missing support narrows the impact claim; it does not prevent arithmetic comparison or a conditional action recommendation.

TECHNICAL APPENDIX

Historical FX is tied to the original shipment month

| Month   | PLN per EUR | CZK per EUR | Published days each |
|---------|-------------|-------------|---------------------|
| 2025-01 | 4.24665455  | 25.16331818 | 22                  |
| 2025-02 | 4.17222500  | 25.07665000 | 20                  |
| 2025-03 | 4.18200952  | 25.00123810 | 21                  |
| 2025-04 | 4.26522000  | 25.03895000 | 20                  |
| 2025-05 | 4.25380000  | 24.92285714 | 21                  |
| 2025-06 | 4.26577143  | 24.80428571 | 21                  |
| 2025-07 | 4.25414783  | 24.62460870 | 23                  |
| 2025-08 | 4.26130476  | 24.51661905 | 21                  |
| 2025-09 | 4.25888636  | 24.34677273 | 22                  |
| 2025-10 | 4.24879130  | 24.31539130 | 23                  |
| 2025-11 | 4.23763500  | 24.23405000 | 20                  |
| 2025-12 | 4.22385714  | 24.25928571 | 21                  |

Arithmetic means use all available published business-day observations in each 2025 month. EUR = 1. No inverse quote, current rate or early rounding is used. JSON stores full numeric means; the audit CSV preserves Decimal precision and the workbook includes all 510 currency-date observations. Sources: [S02](#), [S03](#), [S04](#).



A falling local-per-EUR quote means each local currency unit translates into more EUR, holding price constant. The chart is an indexed historical comparison; the model uses actual monthly means. It does not forecast FX or estimate actual realized conversion gains.

## SOURCE LINKS AND COMPLETION BOUNDARIES

## Evidence register and reproducible handover

**S02 • data-dictionary.md** — Synthetic client source. [Saved source file](#).

**S03 • ecb-history.csv** — Archived official public data. [Original publisher endpoint](#). ECB quote: local currency units per EUR; model period 2025.

**S04 • ecb-history.zip** — Archived official public data. [Original publisher endpoint](#). ECB quote: local currency units per EUR; model period 2025.

**S05 • gdp-per-capita.json** — Archived official public data. [Original publisher endpoint](#). World Bank lastupdated 2026-07-13; 2022–2024 observations.

**S06 • hub-options.csv** — Synthetic client source. [Saved source file](#).

**S07 • order-corrections.csv** — Synthetic client source. [Saved source file](#).

**S08 • orders-part1.csv** — Synthetic client source. [Saved source file](#).

**S09 • orders-part2.csv** — Synthetic client source. [Saved source file](#).

**S10 • population.json** — Archived official public data. [Original publisher endpoint](#). World Bank lastupdated 2026-07-13; 2022–2024 observations.

**S11 • returns.csv** — Synthetic client source. [Saved source file](#).

**S12 • scenario-policy.md** — Synthetic client source. [Saved source file](#).

**S13 • source-register.json** — Archive metadata. [Saved source file](#).

**S14 • unit-costs.csv** — Synthetic client source. [Saved source file](#).

**Provenance.** All 13 indexed source files plus the index were collected from the frozen local source room. Public archive retrieval was 2026-09-27 06:36 UTC; analyst collection was 2026-09-27 10:12 UTC. The full register in evidence/source-register.csv and .json records collection URL, original URL, time, SHA-256, units, period, status and source vintage for every file. Public hashes match the supplied archive register; the ZIP member is byte-identical to the saved ECB CSV. No live source refresh or customer-account access was used.

**Files.** meridian\_analysis.xlsx contains formula-linked financial summaries and scenarios, market context, source/quality notes and charts. metrics.json is the deterministic data exchange. audit/ retains order calculations, exclusions, revision handling, FX observations, scenario runs and thresholds. scripts/ contains source collection, calculation, artifact generation and verification; README.md gives offline reproduction commands.

**Verification.** See verification.json and VERIFICATION.md for actual raw-data cross-checks, workbook cache/formula inspection, PDF rendering and identified limits. Verification of this package does not validate the synthetic commercial assumptions. The 2025 shipment cohort is reproducible; customer demand, saving scope, cash realization, working capital, site feasibility and ramp remain decision gaps.

Public method grounds are documented in decision\_basis.json. Native method selection was available for the portfolio comparison; native activation/application remains unconfirmed. Evidence calibration used a complete installed reference after the initial selector requests failed. No private reasoning or customer interviews are included.